MIL-N-55145B
26 September 1968
Superseding
MIL-W-55145A
30 June 1964

MILITARY SPECIFICATION
NIGHT VISION SIGHT, INDIVIDUAL SERVED WEAPON
AN/PAS-4A

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a near-infrared telescope and light source combination which can be mounted on individual served infantry weapons.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-1/1049	- Electron Tube, Type 6914
MIL-B-18	- Batteries, Dry
MIL-T-152	- Treatment, Moisture-and-Fungus Resistant, of Communications, Electronic and Associated Electrical Equipment
MIL-C-4150	- Case, Carrying, Water-Vaporproof

FSC 5855

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:28Z
Check the source to verify that this is the current version before use.

MIL-P-45536	- Power Supply, Transistorized, 16 Kilovolt
MIL-B-55118	- Batteries, Storage, Vented, Nickel Cadmium
MIL-E-55301	- Electromagnetic Compatibility

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U. S. Military Property

DRAWINGS

SC-DL-637450	- Data List, Night Vision Sight, Individual Served Weapon

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. - The night vision sight, individual served weapon, hereinafter referred to as "weaponsight", shall consist of a viewer, spare parts, manual and a carrying case. The viewer shall consist of a near-infrared-sensitive, image forming telescope with an attached reticle projector, an integral high voltage power supply and battery, a type 6914 electron tube, a near infrared light source, a nickel-cadmium battery and canvas bag, a connecting cable and a weapon adaptor bracket. The weaponsight shall be in accordance with Data List SC-DL-637450 and as specified herein.

3.1.1 Drawings. - The drawings forming a part of this specification are engineering design drawings. Where tolerances prescribed could cumulatively result in improper fits, the contractor shall provide tolerances within those prescribed on the drawings to insure proper fit, assembly, and operation of the weaponsight. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 202640-22T22:28Z
Check the source to verify that this is the currentversion before use.

3.2 Preproduction Models. - As soon as practicable after award of contract, the contractor shall furnish three weaponsights to prove, prior to starting production, that his production methods and choice of design detail will produce weaponsights that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations of weaponsights from the preproduction models shall be subject to the approval of the contracting officer. Approval of the preproduction models by the activity concerned shall not relieve the contractor of his obligation to furnish weaponsights conforming to this specification.

3.3 Insulating and Impregnating Compounds. - Insulating materials and impregnating compounds shall be nonflame-propogating and self extinguising. Impregnating compounds shall not lessen the dielectric strength of or have other injurious effects on the insulation. Impregnating compounds shall not cause corrosion or deterioration of adjacent metal or plastic parts, either on original application or as the result of aging.

3.4 Fungus Resistance. - The weaponsight (except for batteries, the canvas bag, the carrying case, electrical components such as switches, electrical contacts, lamps, reflecting surfaces, radiation transmitting elements, and components hermetically sealed) shall be treated for fungus resistance in accordance with MIL-T-152. The canvas bag shall be treated as specified on the applicable drawings.

3.5 Electromagnetic Compatibility. - The electromagnetic compatibility of the weaponsight shall be in accordance with MIL-E-55301, Tactical Support Class.

3.6 Temperature and Humidity. - The weaponsight shall operate as specified herein and shall not be damaged (see 6.4.1) by operation and storage under the following conditions:

3.6.1 OPERATION

(a) Temperature shocks from plus 125° F to plus 70° F plus or minus 5° F and from minus 25° F to plus 70° F plus or minus 5° F.

(b) Ambient temperatures from plus 115° F to minus 25° F at all relative humidities possible at these temperatures.

(c) A dc voltage from an external source may be supplied to the high-voltage power pack at temperatures below 20° F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:28Z
Check the source to verify that this is the current version before use.

3.6.2 STORAGE

(a) Ambient temperatures from plus 155° F to minus 65° F at all relative humidities possible at these temperatures.

(b) Rapid temperature changes within the range of plus 155° F to minus 65° F with humidity uncontrolled.

3.7 Immersion. - The viewer shall be watertight, shall operate as specified herein, and shall not be damaged after immersion to a depth of 3 feet in fresh water for a period of 30 minutes.

3.8 Resistance to Weapon-Firing Shocks. - The viewer mounted on an M-14, 7.62mm rifle shall not be damaged (see 6.4.2) by the firing of 1,000 rounds from that weapon.

3.9 Viewer

3.9.1 Performance. - Performance at any temperature between 65° F and 75° F shall be as follows:

Resolution, at a maximum of 16,000 volts	Minimum of 0.85 minute for a high-contrast line pattern as shown in figure 1.
Brightness gain, at a maximum of 16,000 volts	Minimum of 0.4
Magnification	4.5 plus 0.5, minus 0.3
Linear Distortion	Not greater than 8 percent at 3 degrees from center of field of view.
Field of View	Minimum of 8 degrees.

3.9.2 High-Voltage Power Supply. - The high-voltage power supply shall conform to MIL-P-45536.

3.9.3 Image Tubes. - The image tube shall conform to MIL-E-1/1049, type 6914.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:28Z
Check the source to verify that this is the current version before use.

3.9.4 Operating Switch. - The switch shall be provided with four operational positions as follows:

a. Full counterclockwise position - OFF.

b. 2nd Position - Telescope ON, reticle OFF.

c. 3rd Position - Telescope ON reticle ON, (low).

d. Full-clockwise position - Telescope ON, reticle ON, (high).

3.9.5 Reticle.

3.9.5.1 Range of Adjustment. - The reticle shall be adjustable over not less than 5 degrees of the telescope field both in azimuth and in elevation. The center of the adjustment range shall be coincident with the optical axis of the telescope. Each click movement in either direction shall change the aiming point 6 inches plus or minus 1 inch at a distance of 300 yards from the telescope.

3.9.5.2 Reticle Intensity. - In the full-clockwise position of the operating switch (see 3.9.4) the reticle pattern shall be visible through the telescope with the telescope aimed at a white-background brightness of 0.1 apparent footlambert as measured from the position of the telescope objective.

3.10 Infrared Light Source.

3.10.1 Beam Characteristics. - With 6 volts applied to the lamp, beam characteristics of the infrared light source without infrared filter shall be as follows:

(a)	Peak beam Candlepower	- Not less than 125,000
(b)	Horizontal Beam Spread	- Not less than 3.5 degrees between the points having an intensity 10 percent of the peak-beam candlepower.

3.10.2 Infrared Filter. - The infrared filter shall have a cut on at 0.8 micron and a minimum of 60 percent transmission at 1.18 microns.

Source: https://assist.dla.mil -- Downloaded: 202640-22T22:28Z
Check the source to verify that this is the current version before use.

3.10.3 Alinement. - The infrared light source shall be mounted on the telescope so that its optical axis and the optical axis of the telescope are parallel. This relationship of the optical axes shall not be change by more than four filament coils in any direction after the firing test.

3.10.4 Reflector. - The quality of the reflecting surface shall be smooth, uniform, and mirror-like in finish and shall be free from surface defects such as:

(a) Cracks.

(b) Scratches.

(c) Holes and Pits.

(d) Flaking.

(e) Orange Peel.

(f) Area with no film.

(g) Fingermarks, dirty or gritty surface.

(h) Dull or distorted areas.

The reflector coating process shall be controlled to yield optimum reflectance of infrared radiation 1 micron in wavelength. The final silicon oxide coating shall be one-half wavelength in optical thickness at 1 micron. The reflector shall be fabricated using the parabolic reflector blank shown on figure 2.

3.11 Batteries. - A BA-42 dry cell conforming to MIL-B-18 shall be used for the power supply and reticle lamp. A BB-429/U nickel-cadmium battery conforming to MIL-B-55118 shall be used for the infrared light source. Unless otherwise specified (see 6.2) the batteries shall be furnished by the contractor.

3.12 Carrying Case. - The carrying case shall conform to MIL-C-4150 except as shown on the drawings.

3.13 Technical Publications. - Such technical publications as are specified shall be furnished (see 6.2).

3.14 Repair Parts and Maintenance Tools. - Such repair parts and maintenance tools as are specified shall be furnished (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:28Z
Check the source to verify that this is the current version before use.

3.15 Items to be Furnished with Weaponsight. - No shipment of weaponsight shall be made unless such repair parts, maintenance tools, operating tools, technical publications, and accessories which are to be furnished with each weaponsight are included with the shipment or unless approval for shipment of the weaponsight without such items has been received from the contracting officer.

3.16 Identification Markings. - Identification markings shall be in accordance with the applicable drawings and MIL-STD-130.

3.17 Government Loaned Property. - The following property in the quantities indicated will be loaned by the Government (see 6.3).

ITEM NO.	DESCRIPTION	FEDERAL STOCK NO.	QUANTITY
1	Rifle, 7.62mm M-14 W/E	1005-589-1271	3
2	Cartridge 7.62mm	1305-301-1676	as required (see 6.3)
3	Pin, Firing	1005-962-8344	3

3.18 Workmanship. - Joints and seams shall be tight fit; all edges shall be smoothly finished. The interior of the infrared light source and telescope shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. - Inspection shall be classified as follows:

a. Preproduction Inspection (see 4.3).

b. Quality Conformance Inspection (see 4.4).

c. Inspection of Preparation for Delivery (see 4.6).

Source: https://assist.dla.mil Downloaded: 2026-10-22T22:28Z
Check the source to verify that this is the current version before use.

4.3 Preproduction Inspection.

4.3.1 Examination. - Each weaponsight shall be examined for the following defects. One or more defects shall be cause for rejection.

MAJOR

101. Broken optical or mechanical components.

102. Does not operate (no characteristic green glow when turned on).

103. Boresight adjustment knob binds.

104. External hardware on viewer loose.

MINOR

201. Technical publications missing or not as specified.

202. Repair parts missing.

4.3.2 Tests. - Upon successful completion of the examination specified in 4.3.1 the preproduction models shall be subjected to the tests in Table I. Tests shall be conducted in the order listed. Failure of any test shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:28Z
Check the source to verify that this is the current version before use.

TABLE I PREPRODUCTION TEST SCHEDULE

TEST	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Firing	4.5.1	3.8
Resolution, Brightness Gain, Magnification	4.5.2	3.9.1
Linear Distortion	4.5.3	3.9.1
Field of View	4.5.4	3.9.1
Range of Reticle Adjustment	4.5.5	3.9.5.1
Click Movement	4.5.6	3.9.5.1
Reticle Intensity	4.5.7	3.9.5.2
Infrared Light Source Peak Beam Candlepower and Beam Spread	4.5.8	3.10.1
Infrared Light Source Alinement	4.5.9	3.10.3
Humidity	4.5.10	3.6
Low and High Temperature	4.5.11	3.6
Immersion	4.5.12	3.7
Infrared Filter	4.5.13	3.10.2
Electromagnetic Compatibility	4.5.14	3.5
Resolution, Brightness Gain, and Magnification	4.5.2	3.9.1

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:28Z
Check the source to verify that this is the current version before use.

4.4 Quality Conformance Inspection

4.4.1 Examination. - Each weaponsight shall be examined as specified in 4.3.1. Presence of one or more defects shall be cause for rejection of that unit.

4.4.2 Tests.

4.4.2.1 Group A. - Each weaponsight shall be tested in accordance with Table II. Failure of any test shall be cause for rejection of that unit.

TABLE II GROUP "A" TEST SCHEDULE

TEST	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Resolution, Brightness Gain and Magnification	4.5.2	3.9.1
Infrared Light Source Peak Beam Candlepower and Beam Spread	4.5.8	3.10.1

4.4.2.2 Group B. - Group B tests shall be conducted on weaponsights selected from units which have passed the examination and tests in 4.4.1 and 4.4.2.1. Sampling shall be in accordance with MIL-STD-105, AQL 1.5 percent defective, inspection level S-3. Tests shall be conducted in accordance with Table III. Failure of any test shall be cause for rejection.

TABLE III GROUP "B" TEST SCHEDULE

TEST	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Immersion	4.5.12	3.7
Resolution, Brightness Gain, and Magnification	4.5.2	3.9.1

Source: https://assist.dla.mil -- Downloaded: 202640-22T22:28Z
Check the source to verify that this is the current version before use.

4.4.2.3 Group C. - This inspection shall consist of the tests specified in Table IV and shall be performed on sample units that have been subjected to and met Group A and B inspection. Sample units shall be selected in accordance with 4.4.2.3.1. The tests may be performed in any order except the resolution, brightness gain and magnification test shall be last.

4.4.2.3.1 Sampling for Inspection. - Three sample weaponsights shall be selected at random for all inspections in Table IV from the monthly specified quantity (see 6.2) produced or every thirty (30) days, whichever occurs first. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

TABLE IV GROUP "C" TEST SCHEDULE

TEST	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Firing	4.5.1	3.8
Humidity	4.5.10	3.6
Low and High Temperature	4.5.11	3.6
Infrared Filter	4.5.13	3.10.2
Click Movement	4.5.6	3.9.5.1
Field of View	4.5.4	3.9.1
Linear Distortion	4.5.3	3.9.1
Range of Reticle Adjustment	4.5.5	3.9.5.1
Reticle Intensity	4.5.7	3.9.5.2
Infrared Light Source Alinement	4.5.9	3.10.3
Resolution, Brightness Gain, and Magnification	4.5.2	3.9.1

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:28Z
Check the source to verify that this is the current version before use.

4.4.2.3.2 Noncompliance. - The contractor shall immediately report in writing each Group C failure occurence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.4.3 Reinspection of Conforming Group B and Group C Sample Units. - Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5 Test Procedure. - Unless otherwise specified all tests except firing test shall be performed at plus 70° F, plus or minus 5° F.

4.5.1 Firing Test. - The test may be conducted during daylight hours. The number of rounds to be fired shall be as follows:

a.	For Preproduction Testing	1000 rounds per sight
b.	For Quality Conformance Testing	500 rounds per sight

The viewer shall be mounted on the M-14 rifle and shall be sighted in on target. Sighting in shall be as follows: First the weapon with the viewer attached thereon shall be mounted on a rigid holding device with the viewer focused and sighted at a target 200 yards away ± 10 percent. Using the normal open sights of the weapon, adjust for the range. Next the viewer boresight shall be adjusted until the aiming reference dot coincides with the center of the crosshair of the target. All hardware and connectors shall be checked to insure that no parts are loose. The weapon and viewer shall be removed from the holding device and 200 rounds

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:28Z
Check the source to verify that this is the current version before use.

shall be fired from either the prone position or the off-hand position. During and after the firing of 200 rounds, observe and record any internal flashing, flickering, loosening or failure of any hardware or adjustments. After 200 rounds are fired, the weapon and viewer shall be replaced in the holding device and the position of the reticle with respect to the original settings shall be determined. Any change in boresight or reticle shift, or any other evidence of failure to conform to the requirements of this specification shall be recorded. The 200 rounds firing and examination cycle shall be repeated as often as required to accumulate the total number of rounds specified to be fired. No adjustments, tightening of parts, or repairs of any kind shall be permitted after the initial boresighting is completed and the test firing cycles have begun without proper recording of same. At the conclusion of the test, examine the viewer for damage as defined in 6,4.2. Evidence of damage shall constitute failure of this test.

4.5.2 Resolution, Brightness Gain and Magnification. - These tests shall be made in an ambient temperature between 65° F and 75° F and with a voltage of not more than 16,000 volts applied to the image tube by the power supply.

4.5.2.1 Resolution and Brightness Gain. - The resolution and brightness-gain tests shall be made on the testing apparatus shown in figure 1 or other equivalent test apparatus. The testing apparatus with the telescope removed and with the resolution chart shown in figure 3 and the neutral density filter in place shall be calibrated by adjusting the oscilloscope to obtain the waveform shown in figure 4. The neutral filter shall then be replaced by the infrared filter and the viewer shall be placed in the testing apparatus and centered with the optical axis of the testing apparatus. The waveform shown in figure 5 should then be obtained on the oscilloscope. Measure B, C, and D indicated on figure 5. Results other than C/D equal to 0.3 (which corresponds to an angular resolution of 0.85 minute) or larger; B equal to 0.4 (which is obtained when the objective transmission is 0.72 and its aperture ratio is f/3, the eyepiece transmission is 0.82, and the image-tube magnification and conversion index are 0.75 and 15, respectively) or larger shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:28Z
Check the source to verify that this is the current version before use.

4.5.2.2 Magnification. - The resolution chart from the test instrument in figure 1 shall be removed and replaced with a target having a pattern of 1-degree width. The photo tube shall be removed and replaced with a white diffusing screen. The width of the projected pattern on the screen shall be measured without the telescope in the instrument. The telescope shall be placed in position and either the screen or the telescope shall be adjusted for a sharp image on the screen. The width of the projected pattern shall be measured. An image other than 4.5 plus 0.5 or minus 0.3 times larger than the first projected image shall constitute failure of this test.

4.5.3 Linear Distortion. - With the telescope in place on axis as shown in figure 6, both targets will be projected on the half-silvered mirror; target No. 1 will be projected through the telescope. The circle of the pattern of target No. 1 not falling between the two circles of the pattern of target No. 2 shall constitute failure of this test.

4.5.4 Field of View. - The half-silvered mirror shall be removed from the testing apparatus used in 4.5.3. The telescope shall then be pivoted either to left or right until the center point of the cross on target No. 1 is at the edge of the field of view. The telescope shall then be pivoted in the opposite direction until the center point of the cross is at the opposite edge of the field of view. The total angle through which the telescope was pivoted shall be measured. This angle is defined as the field of view of the telescope. A field of view of less than 8 degrees shall constitute failure of this test.

4.5.5 Range of Reticle Adjustment. - Use the test instrument specified in 4.5.3 with a target consisting of a circle subtending 5 degrees of the telescope field of view and having a center mark placed in the instrument. The target shall be concentric with the optical axis of the telescope. The reticle shall be adjustable to the extremities of this circle. Failure of the reticle to attain this adjustment shall constitute failure of this test.

4.5.6 Click Movement. - Mount the telescope in a rigid holding position. With the image tube and reticle lamp on, view a target at a distance of 25 yards. Mark the position of the projected reticle image on the target. Turn the reticle azimuth adjustment 24 clicks clockwise or counterclockwise. Measure the displacement of the projected reticle image on the target. The above procedure shall be repeated in elevation. Reticle azimuth or elevation adjustment of less than 10 inches or more than 14 inches shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:28Z
Check the source to verify that this is the current version before use.

4.5.7 Reticle Intensity. - Remove the diffuse screen, resolution chart, and filters from the instrument shown in figure 1. Place a diffusing screen of such transmission that a brightness of 0.1 foot-lambert is presented to the telescope objective in the position occupied by the resolution chart. Remove the photo tube, vertical slit, and pickup lens and view the diffusing screen through the telescope. A reticle pattern not visible with the switch on the full-clockwise position shall constitute failure of this test.

4.5.8 Infrared Light Source Peak Beam Candlepower and Beam Spread. - Measure the infrared light source and beam spread without infrared filter 100 feet from the light source. The receiver may be a viscor-filtered photo-electric cell and a low-internal-resistance microammeter; a recorder calibrated to read footcandles; or a viscor-filtered foot-candle meter. Before tests are made, any receiver used shall be calibrated against a National Bureau of Standards (NBS) standard lamp with a color temperature of 2870° K. To measure the peak-beam candlepower, move the receiver through the light-source beam or move the light-source beam over the receiver, both vertically and horizontally, until the peak foot-candle reading is obtained. The formula for converting this reading to candlepower is:

$$I = Ed^2$$

WHERE I = Candlepower of the light source

E = Foot-candles measured by the receiver

d = Distance from the light source to the receiver

A minimum peak candlepower of less than 125,000 shall constitute failure of this test. To determine the beam spread, place the receiver in a stationary position and rotate the light source until a peak-beam candlepower of not less than 125,000 is obtained on the receiver. Rotate the light source to the left until the receiver indicated 10 percent of the peak-beam candlepower. Then rotate the light source to the right until the receiver again indicates 10 percent of the peak-beam candlepower. Measure and record the angle through which the light source is rotated between the 10 percent points of peak-beam candlepower. A minimum peak candlepower of less than 125,000 or a horizontal beam spread of less than 3.5 degrees between points having an intensity of 10 percent of the peak-beam candlepower shall constitute failure of this test. All light sources will be focused for 200 yards.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:28Z
Check the source to verify that this is the current version before use.

4.5.9 Infrared Light Source Alinement. - Place the telescope in the instrument used in 4.5.2.1 and adjust the reticle to the center position. Replace the collimator and the light source of the instrument with either a reflective or refractive collimator not less than 12 inches in diameter. Place a target at the focal point of the collimator. Turn on the infrared light source and the telescope and view the target through the telescope. Determine the position of the light source filament image with respect to the reticle. The aiming point of the reticle and the center of the light source filament image must coincide for parallel adjustment. This relationship of the optical axes shall not be changed by more than four filament coils in any direction after the firing test. Failure of the aiming point of the reticle and the center of the filament image to coincide shall constitute failure of this test.

4.5.10 Humidity. - Place the viewer in a chamber with an internal temperature of plus 85° F and a relative humidity of 90 to 98 percent and soak for 4 hours. Raise the temperature of the chamber to 120° F in not more than 2 hours, maintain the relative humidity of 90 to 98 percent and soak for 4 hours. This constitutes one complete cycle as shown in figure 7. Perform the above cycle six times. Turn the viewer on and observe for tube operation for the last 30 minutes of the 85° F portion of the 4th and 6th cycles. Failure to operate or evidence of damage during, or immediately following, the above six cycles shall constitute failure of this test.

4.5.11 Thermal Shock, Operation, High and Low Temperature Storage. - Place the viewer in a chamber which has an internal temperature of 155° F, a relative humidity of 20 to 25 percent and soak for 4 hours. Lower the temperature in the chamber to 115° F in not more than 2 hours and soak for 4 hours. Turn the viewer on and observe for tube operation during the first 20 minutes of this 4 hour period. Remove the viewer from the chamber and soak for 2 hours maintaining relative humidity at 70° F. Place the viewer in a chamber which has an internal temperature of minus 65° F, 100 percent relative humidity. Soak the viewer at this temperature for 12 hours. Turn the viewer on and observe for tube operation for the last 30 minutes of the twelfth hour. It is permissible to replace the battery in the unit at this time, with a battery which has been at 70° F in order to operate this unit. Remove the viewer from the chamber and soak for 4 hours at 70° F uncontrolled humidity. Turn the viewer on and observe for tube operation for the last 10 minutes of this period. This constitutes one complete 28 hour cycle as shown in figure 8. Repeat the above cycle. Failure to operate or evidence of damage during or immediately following the above two cycles shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:28Z
Check the source to verify that this is the current version before use.

4.5.12 Immersion. - Immerse the viewer to a minimum depth of 3 feet in fresh water for a period of 30 minutes. Dry all external surface, disassemble, and inspect for entrance of water or moisture. Any evidence of water or moisture within the viewer shall constitute failure of this test.

4.5.13 Infrared Filter. - Place the filter in a spectrophotometer and perform the following test. Failure of the infrared filter transmittance to fall within the upper and lower limits shall constitute failure of the test.

MICRONS	TRANSMITTANCE	
	LOWER LIMIT	UPPER LIMIT
1.95	80%	None
1.41	75%	None
1.18	60%	None
1.07	45%	None
1.03	32%	None
0.97	5%	None
0.90	None	None
0.80	None	None
0.749	None	5%
0.660	None	9%
0.546	None	20%

4.5.14 Electromagnetic Compatibility. - The preproduction model, equipped for the reduction of radio interference in accordance with 3.5 shall be tested as specified in MIL-E-55301, except that the time interval for submission of reports shall be as specified (see 6.2).

4.6 Inspection of Preparation for Delivery. - The packing and marking shall be examined to determine compliance with Section 5 of this specification.

Source: https://assist.dla.mil -- Downloaded: 202640-22T22:28Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking. - The viewer with spare parts, tools and publications shall be placed in the carrying case. The carrying case clasps shall be closed and two wire and lead tamperproof seals affixed. One seal shall be affixed to each side of the case through a clasp in a manner that the clasps cannot be released without destroying the seal. The carrying case shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use. - The weaponsight is intended for use primarily to permit accurate and visually secure night firing of infantry weapons and secondarily as a countermeasures aid in the night detection of enemy use of infrared light sources and as a general-purpose infrared surveillance telescope.

6.2 Ordering Data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When the contractor will not furnish the batteries (see 3.11).

c. Technical publications to be furnished (see 3.13).

d. List of repair parts and maintenance tools (see 3.14).

e. Monthly delivery rate (see 4.4.2.3.1)

f. Time interval for submission of reports (see 4.5.14).

6.3 The contracting officer should arrange to loan the property specified in 3.17. Quantities of ammunition required will vary with the number of weaponsights selected for quality conformance testing (500 rounds per quality conformance test and 1,000 rounds per pre-production test).

6.4 Definitions. - The following definitions shall apply throughout this specification.

6.4.1 Operation and Storage Damage. - Breakage, loosening, failure, deterioration, or corrosion of any finish, hardware, connection or component. Condensation of moisture within the telescope or light source and high voltage breakdown. Failure to operate in accordance with provisions of this specification.

Source: https://assist.dla.mil -- Downloaded: 202610-22T22:28Z
Check the source to verify that this is the current version before use.

6.4.2 Shock Damage. - Any loosening of parts, change of sighting adjustment, failure or breakage of any hardware, connection, or components; any deterioration in quality of the image; any evidence of intermittent electrical contact; any change in boresight greater than 1 mil; any shift of alinement greater than the equivalent of four filament coils of the lamp.

6.5 International Standardization. - Certain provisions of this specification are the subject of international standardization agreement, ABC STD 121, 128 and 145; ABCA STD 146 and NATO STANAG 2324, 2325, 2330 and 2331, as indicated on the drawings. When amendment, revision, or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

CUSTODIAN:
ARMY - EL

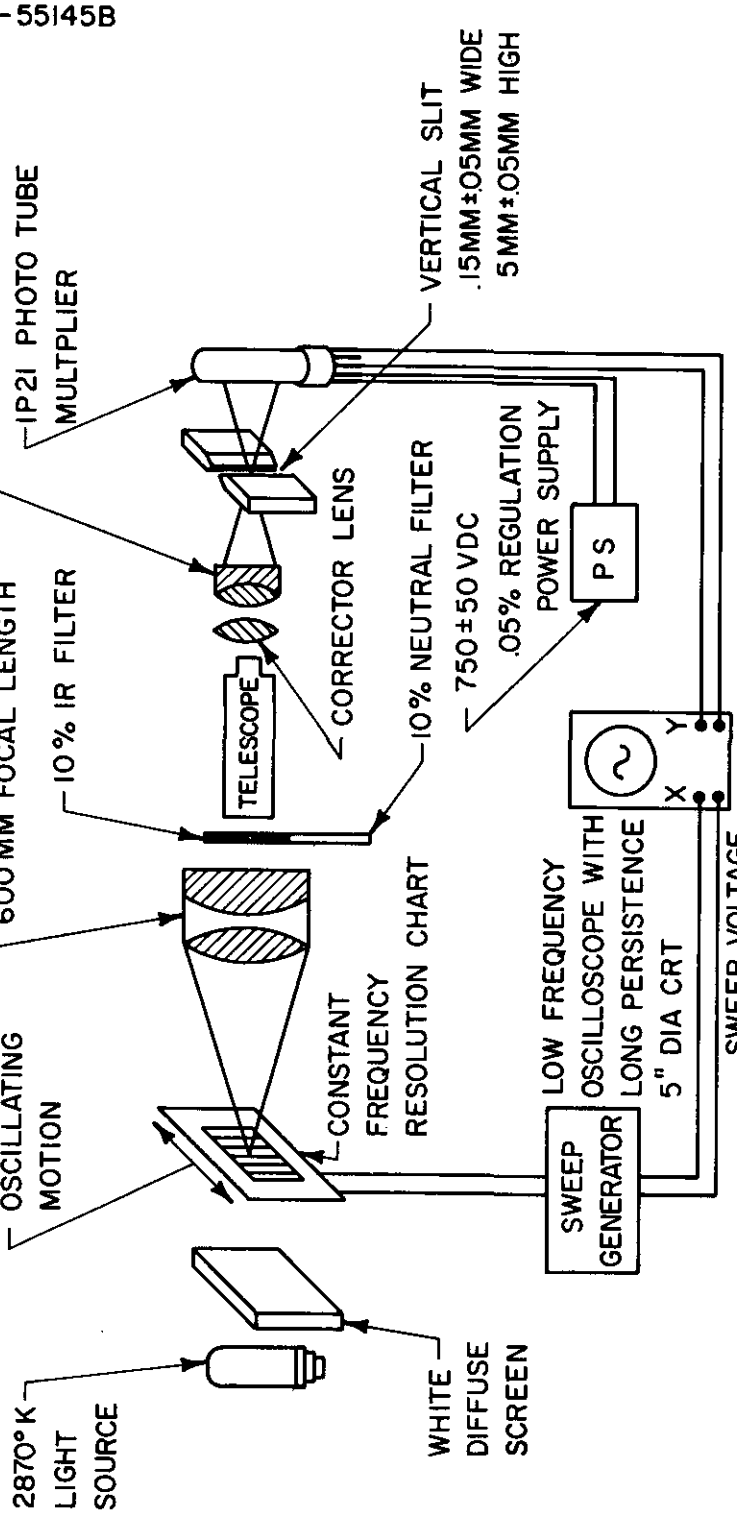
PREPARING ACTIVITY:
ARMY - EL

REVIEW ACTIVITIES:
ARMY - WC

USER ACTIVITIES:
ARMY - MD, MU, GL
Navy - AS, EC, MC

PROJECT NO. 5855-0001

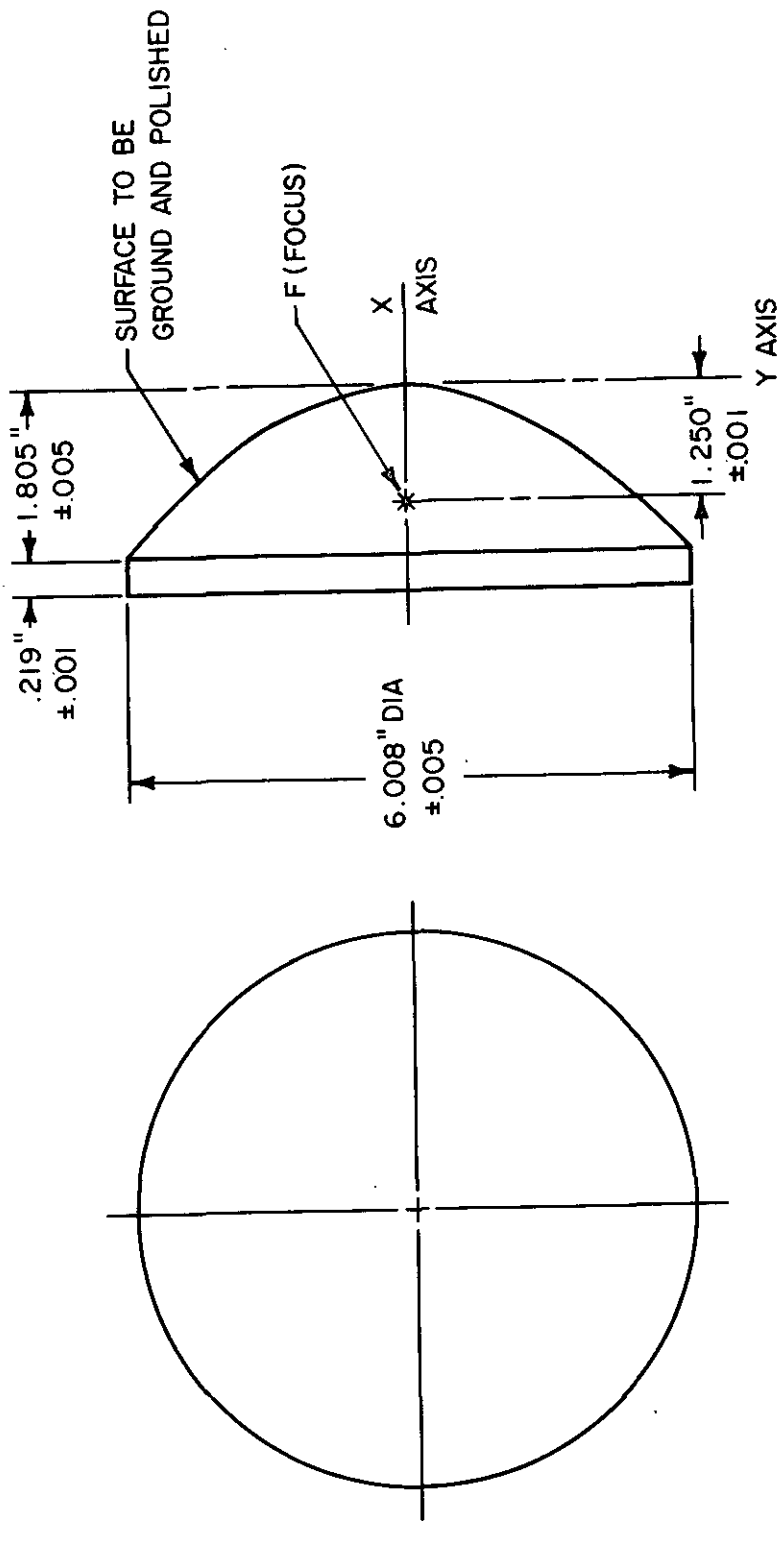
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:28Z
Check the source to verify that this is the current version before use.



NOTES:
1. LOCATE PICKUP LENS 9" ±1" FROM VERTICAL SLIT.
2. COLLIMATOR AND SLIT TO BE CENTERED AS REQUIRED.

RESOLUTION TEST SYSTEM
FIGURE 1

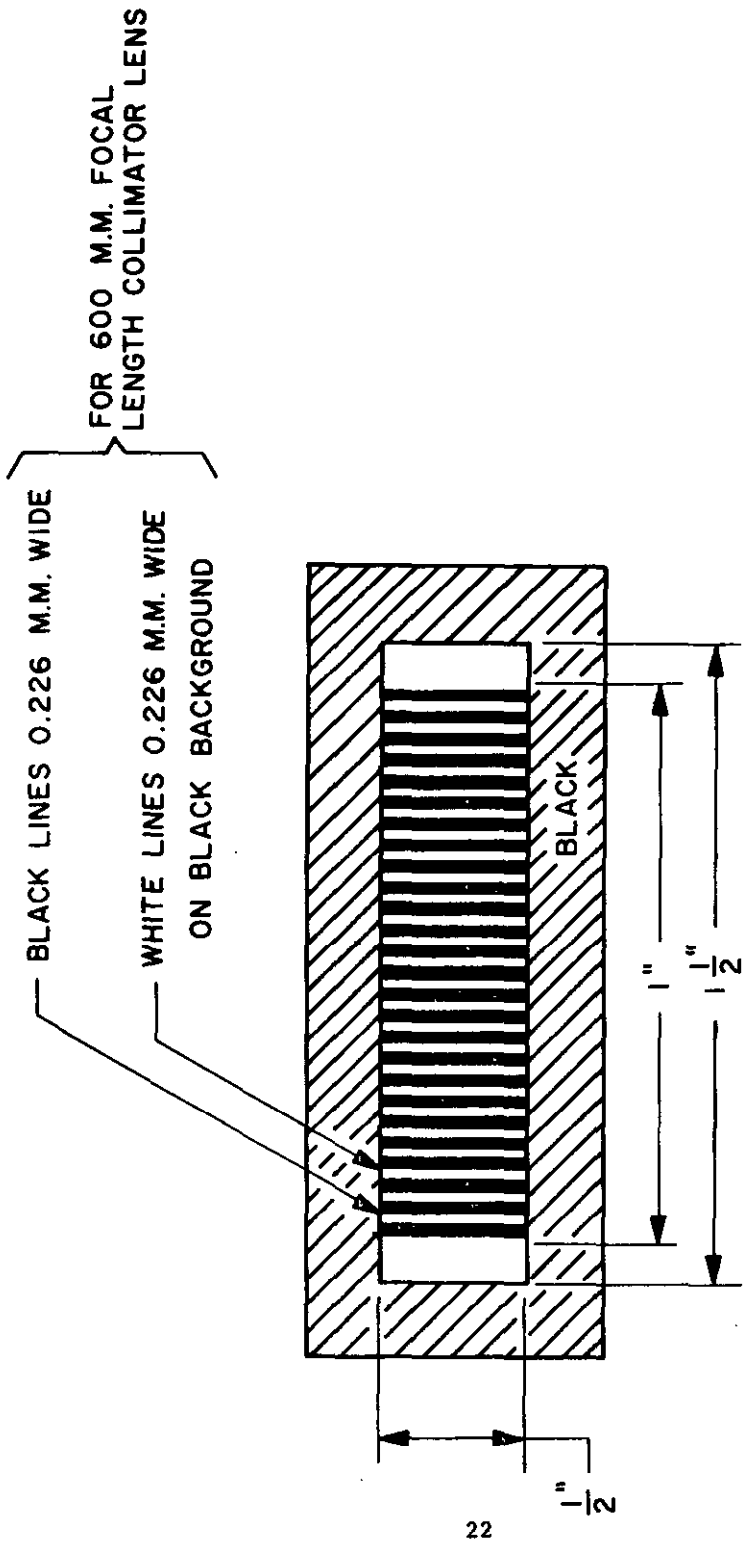
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:28Z
Check the source to verify that this is the current version before use.



NOTES:
1. PARABOLIC SHAPE DERIVED FROM THE FORMULA $X = \frac{Y^2}{4F}$.
2. MATERIAL- PYREX GLASS #7740.

PARABOLIC REFLECTOR BLANK
FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:28Z
Check the source to verify that this is the current version before use.

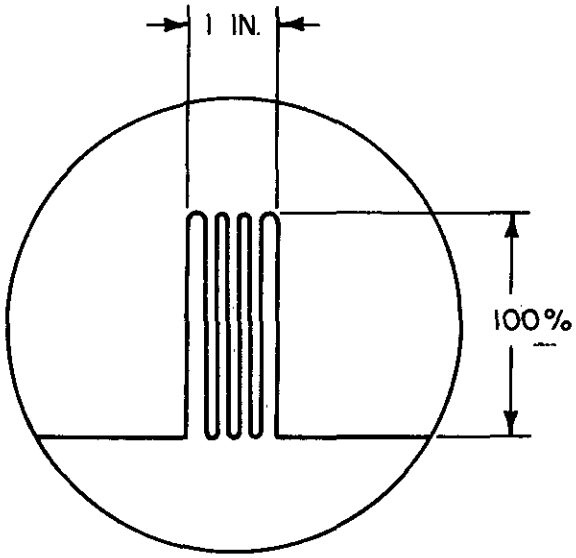


NOTE

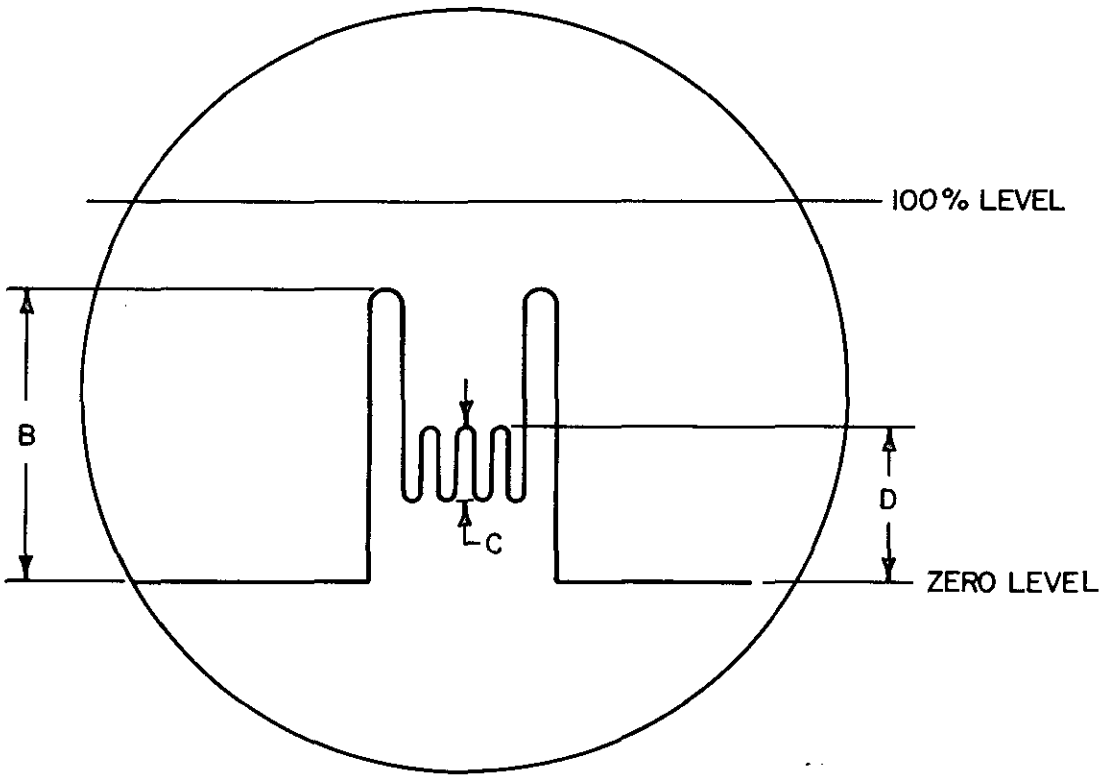
HIGH CONTRAST CHART≈50/1-100/1

RESOLUTION CHART
FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:28Z
Check the source to verify that this is the current version before use.

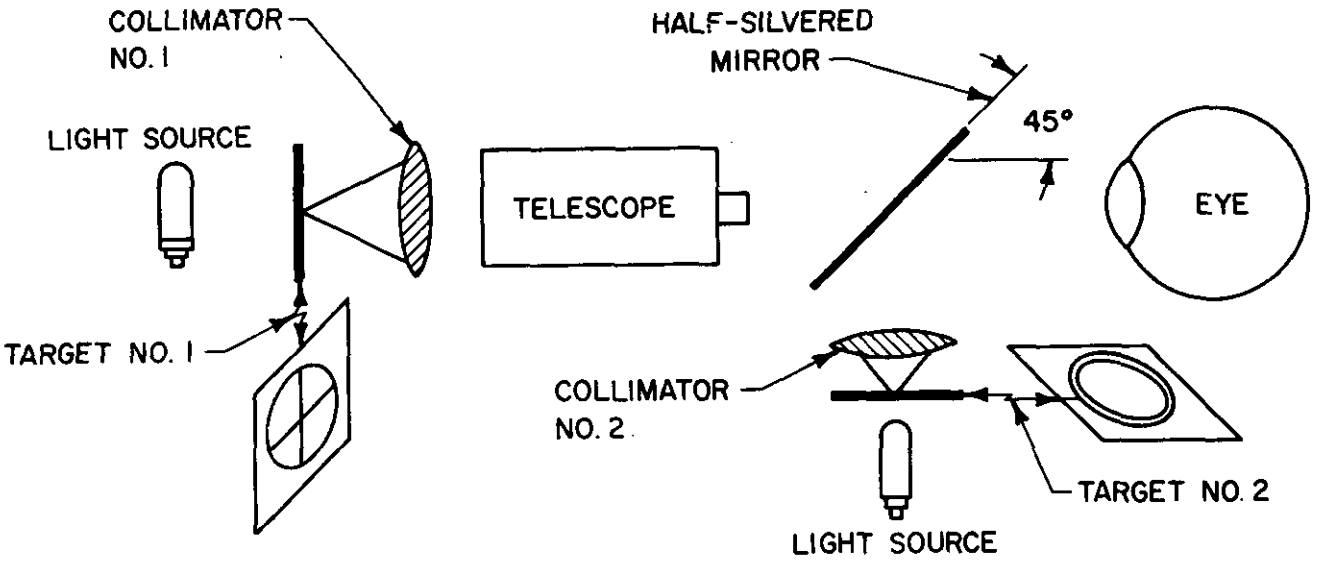


WAVEFORM, NEUTRAL FILTER
FIGURE 4



WAVEFORM, INFRARED FILTER AND TELESCOPE
FIGURE 5

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:28Z
Check the source to verify that this is the current version before use.



NOTES:

A. COLLIMATOR NO. 1 AND COLLIMATOR NO. 2 SHALL BE IDENTICAL.
B. TARGET NO. 2 SMALL RING SHALL BE 4.4 TIMES LARGER THAN TARGET NO. 1.
C. TARGET NO. 2 LARGE RING SHALL BE 5.07 TIMES LARGER THAN TARGET NO. 1.

LINEAR DISTORTION TEST SYSTEM
FIGURE 6

Source: https://assist.dla.mil -- Downloaded: 2026-02-22T22:28Z
Check the source to verify that this is the current version before use.

RELATIVE HUMIDITY TEST CYCLES

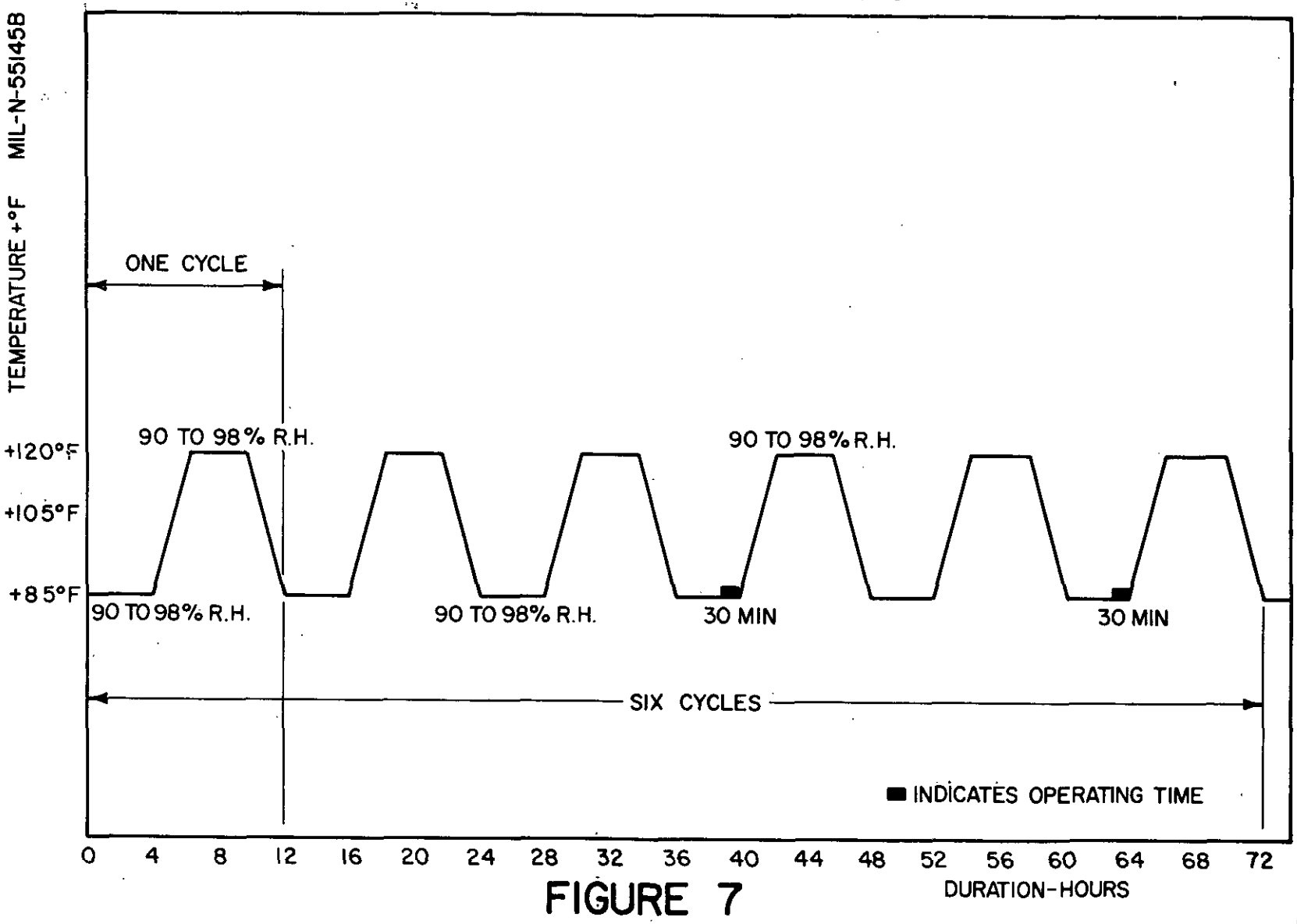


FIGURE 7

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:28Z
Check the source to verify that this is the current version before use.

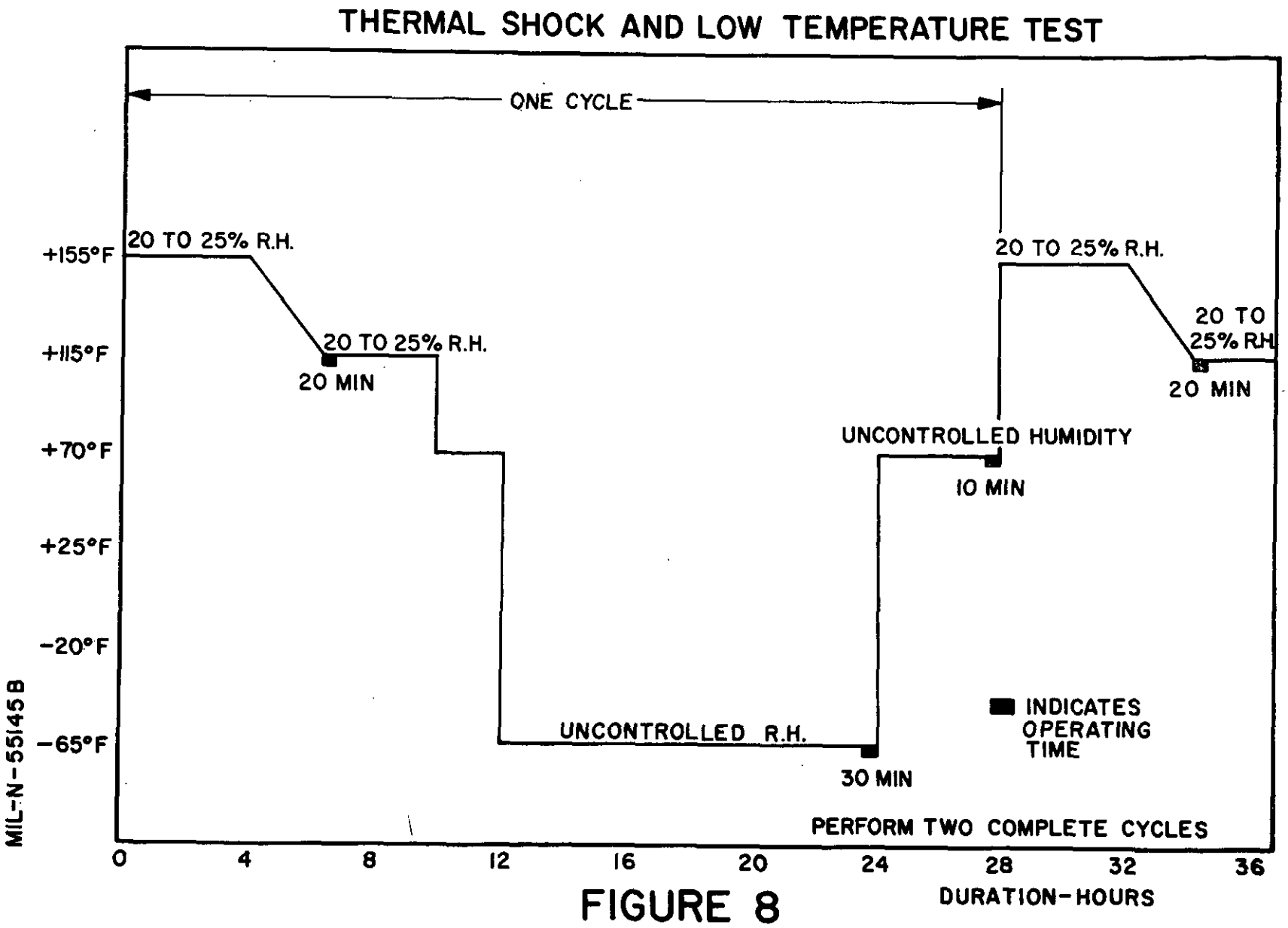


FIGURE 8

Source: https://assist.dla.mil -- Downloaded: 202640-22T22:28Z
Check the source to verify that this is the current version before use.